Exercise Sheet 4

Centrality and Structural Roles

5942UE Network Science

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4.A Centrality Measures

Exercise 4.A.1: Centrality by Hand

Consider the graph with nodes A, B, C, D, E, F and edges A–B, A–C, B–C, B–D, D–E, D–F, E–F.

1. Compute **degree centrality** $C_d(v) = \deg(v)/(|V| - 1)$ for all nodes.
2. Identify the shortest paths between all node pairs. Which node lies on the most shortest paths between other nodes? (Compute betweenness centrality by inspection.)
3. Which node has the highest degree centrality? Which has the highest betweenness centrality? Are they the same node?
4. Interpret: what structural role does the node with the highest betweenness play in this network?

Solution:

1. Degree centrality ($|V| - 1 = 5$):

Node	Degree	C_d
A	2	2/5
B	3	3/5
C	2	2/5
D	3	3/5
E	2	2/5
F	2	2/5

2. Betweenness — by inspection:

The graph has two triangle-like clusters: A, B, C on the left and D, E, F on the right, joined by edge B–D. Every shortest path from any node in A, C to any node in D, E, F must pass through B

then D. By symmetry, B and D have the same betweenness: each lies on 6 shortest paths between other node pairs (normalised betweenness $= 0.6$).

3. Highest degree centrality: B and D are tied at $3/5$. **Highest betweenness:** B and D are also tied. They are the two endpoints of the bridge edge between the two triangles.

4. Structural role of B and D: Both are **broker** or **bridge** nodes. They sit on the unique paths connecting the left cluster A, B, C to the right cluster D, E, F. Removing either one would disconnect the two groups. High betweenness with moderate degree is the signature of a structural bridge.

Exercise 4.A.2: When Centrality Measures Disagree

Using `nx.karate_club_graph()`:

1. Compute degree, betweenness, closeness, and eigenvector centrality for all nodes.
2. Find the top-3 nodes for each measure. Do the top lists overlap?
3. Plot four subplots, each showing the network with node sizes proportional to one centrality measure.
4. Find a node that ranks **very differently** across measures. What does this reveal about its structural position?

Solution:

```python
import networkx as nx
import matplotlib.pyplot as plt

G = nx.karate_club_graph()

# Centrality dashboard
centralities = {
    "Degree":      nx.degree_centrality(G),
    "Betweenness": nx.betweenness_centrality(G),
    "Closeness":   nx.closeness_centrality(G),
    "Eigenvector": nx.eigenvector_centrality(G),
}

# Top-3 per measure
for name, c in centralities.items():
    top3 = sorted(c, key=c.get, reverse=True)[:3]
    print(f"{name}: {top3}")
```

Interpretation:

- **Leading hubs:** Nodes 0 and 33 lead in almost every measure — they are the faction leaders with high degree and influence.
- **Structural disagreement:** Nodes can rank differently because each centrality encodes a different theory of importance. For example, a node can have high betweenness if it connects regions of the graph, even when it is not one of the strongest hubs by degree or eigenvector centrality.

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4.B Structural Roles

Exercise 4.B.1: Embedded vs. Broker

Consider a graph with two triangles $1, 2, 3$ and $4, 5, 6$, plus node 7 connected only to nodes 3 and 4. There is no direct edge 3–4.

1. Is node 1 **embedded** or a **broker**? What evidence supports your answer?
2. Is node 7 **embedded** or a **broker**? What is its clustering coefficient C_7?
3. If edge 3–4 is added, what happens to node 7's structural role?
4. Burt (1992) argues brokers gain "information advantage." What unique information could node 7 access that nodes in $1, 2, 3$ cannot?

Solution:

1. Node 1 — embedded: Node 1 is surrounded by the closed triangle $1, 2, 3$ where edges 1–2, 1–3, 2–3 all exist. Its neighbours are mutually connected. Clustering coefficient $C_1 = 1$. Node 1 cannot reach $4, 5, 6$ without going through 3, 7, and 4, so it has no direct brokerage opportunity.

2. Node 7 — broker: Node 7 connects to 3 and 4, which are the two contact points for the two triangles. Are 3 and 4 connected? No. So $C_7 = 0$ (the one possible edge between 7's two neighbours is absent). Node 7 bridges a structural hole: it is the only route between the $1, 2, 3$ cluster and the $4, 5, 6$ cluster.

3. If 3–4 is added: Node 7 would no longer be the only route between the two triangles. Its clustering coefficient becomes $C_7 = 1$ because its two neighbours are now connected, and its brokerage role becomes weaker because the graph has a direct alternative path between the two groups.

4. Information advantage: Node 7 can learn about jobs, ideas, and events in the $4, 5, 6$ cluster that are invisible to $1, 2, 3$ (and vice versa), because information circulates within dense clusters but rarely

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Exercise 4.B.2: Betweenness as a Proxy for Brokerage

Using the two-triangle + node 7 graph from Exercise 4.B.1:

1. Build the graph in NetworkX. Compute betweenness centrality for all nodes.
2. Verify that node 7 has high betweenness and low clustering coefficient.
3. Explain why this combination is a good proxy for brokerage.
4. On the karate club graph: find the 3 nodes with the highest betweenness but **lowest** clustering coefficient. What does this combination mean structurally?

Solution:

```python
import networkx as nx

G = nx.Graph()
G.add_edges_from([(1,2),(2,3),(1,3), (4,5),(5,6),(4,6), (7,3),(7,4)])

bc = nx.betweenness_centrality(G, normalized=True)
cc = nx.clustering(G)

print("Node 7 betweenness:", bc[7])
print("Node 7 clustering:", cc[7])

# High betweenness and low clustering: sort by betweenness first,
# then use clustering coefficient to distinguish bridge-like positions.
K = nx.karate_club_graph()
bc_k = nx.betweenness_centrality(K)
cc_k = nx.clustering(K)
top_brokers = sorted(K.nodes(), key=lambda n: (-bc_k[n], cc_k[n]))[:3]

for n in top_brokers:
```

Interpretation:

- **Metric contrast:** Clustering coefficient is a local density measure (neighbours connected?), while betweenness is a global routing measure (on shortest paths?).
- **Brokerage signature:** Node 7 has high betweenness because all shortest paths between the two triangles pass through it, and $C_7 = 0$ because its two neighbours are not directly connected.
- **Pure bridges:** High betweenness with low clustering indicates a bridge-like position: the node connects otherwise separate parts of the graph rather than sitting inside one dense neighbourhood.